Text-to-SQL Models Make Choices Without Saying So [Experiment, Analysis & Benchmark]: Supplementary Material

Yichen Xu
University of California, Berkeley
Berkeley, CA, USA
yichen_xu@berkeley.edu

Shipei Lin
School of Data Science
The Chinese University of Hong Kong, Shenzhen
Shenzhen, China
225045017@link.cuhk.edu.cn

Chenhao Ma
School of Data Science
The Chinese University of Hong Kong, Shenzhen
Shenzhen, China
machenhao@cuhk.edu.cn

This supplement adds evidence and detail to the paper; it does not repeat it. Section, figure, table and equation numbers without an S refer to the paper. Appendix A maps the data and the experiments. Appendix B shows that the choices are common: how often benchmark references meet each type, and on how many databases it occurs. Appendix C lists documented cases outside SQL in which one such choice changed a consequential number. Appendix D reports the request sets whose choice lies in a filter condition. Appendices E and F give the generator's details, the seeds and the versions. Appendix G shows examples of every type, selected by a fixed rule. The data and code behind every number are in the reproducibility package of the repository.

A DATA AND EXPERIMENTS

Figure S1 shows the data of the paper at three levels: the sources, the data built from them, and each experiment with its sample size. Request sets are built on the evaluation databases from generated tasks and from the choice library, which is mined from BIRD and Spider development questions; the held-out questions are not used for mining. Three subsets come from random draws made before the final selection of request sets, and the figure gives each draw with the part of it that lies in the final sets. Each experiment lists the exhibits it supports, so every figure and table of the paper can be traced to its data. The counts are generated from the same data files as the paper's numbers.

B THE CHOICES ARE COMMON

The choices in the library are not quirks of one benchmark or one database. Table S1 lists every type, how often BIRD and Spider references meet it, and how many request sets test it. Table S2 shows on how many databases, and in which sources, each type occurs. Appendix G gives examples of every type that a reader can check by hand.

The library. Table S1 gives the labels and counts in the fixed example library for every type. "Open" includes both unresolved and contested records. "Settled" denotes failure records the reviewers label as settled by the wording; this label is not the word-list check used for the defaults experiment. Failures that differ only in output format (51 with two accepted interpretations, 10 with one) are not choices of meaning and are not in the table. Possible reference errors (27) and unplaced cases (four) are also kept apart. For Table 1, AmbiQT's column and table ambiguity and AMBROSIA's vagueness map to *which column*, AMBROSIA's scope to *any vs. all*, and attachment to *how conditions link*.

Table S1: All choice types (Figure 4 of the paper shows the major ones). Library: choices of meaning in benchmark failures, open (the wording fits both interpretations) or settled (it selects one). References: BIRD and Spider questions silent on the choice, by the interpretation their reference SQL takes: the usual one, or the other with its share (%), after the less common interpretations were reviewed; –: not readable from the SQL. Sets: the 204 accepted request sets, of the types whose other interpretation at least 1% of silent references take. The library's value-representation type is split by its SQL edit: comparison strictness is a strict cutoff, any other edit value matching.

		Library		References		
Choice type	The choice	Open	Settled	Usual	Other (%)	Sets
Aggregation level	Rows or group averages?	6	2	244	15 (5.8)	61
Calculation	Which formula is meant?	1	1	613	1 (0.2)	0
Quantifier	"Than any": one or all?	3	0	8	7 (46.7)	0
Condition linkage	Conditions on one row?	1	0	260	0 (0.0)	0
Counting unit	Count rows or entities?	7	27	2,521	237 (8.6)	30
Missing values	Keep or skip missing?	8	3	1,630	26 (1.6)	29
Negation	Some non-match, or none?	0	1	272	3 (1.1)	1
Population scope	Keep unmatched rows?	0	1	12,462	5 (0.0)	0
Referent column	Which column is meant?	19	6	–	–	0
Relationship role	Which link to follow?	2	4	15	0 (0.0)	0
Strict cutoff	Strict or inclusive?	1	0	309	5 (1.6)	42
Ties and selection	One tied row, or all?	7	3	2,826	397 (12.3)	41
Time scope	Period? Ends included?	1	2	254	1 (0.4)	0
Value matching	Exact or loose match?	1	3	9,330	22 (0.2)	0
Total		57	53			204

How references are counted. The References columns of Table S1 read each reference query with an SQL parser and the schema's keys and column types, without a model, and count a question only when its wording has no cue for the choice. Ties, counting unit, missing values, population scope, "between" and value matching follow Section 6.1 of the paper; calculation is a question that says percent, answered times 100 or as a fraction. *Quantifier*: a question that says "than any", answered by comparing with every value (`> MAX`, `< MIN`, `ALL`) or with at least one (`> MIN`, `< MAX`, `ANY`). *Negation*: a negated question answered with no matching record (`NOT IN`, `NOT EXISTS` or `EXCEPT` over another table) or with some non-matching record (<> on the many side of a join). *Aggregation level*: an entity ranked by a numeric column on the many side of a join, by one row's value or by a sum or average per group. *Condition linkage*: two or more conditions on different columns of the many side of a join, on one row or on separate rows (the table joined twice by the same key, or an `INTERSECT`). *Relationship role*: two tables that share

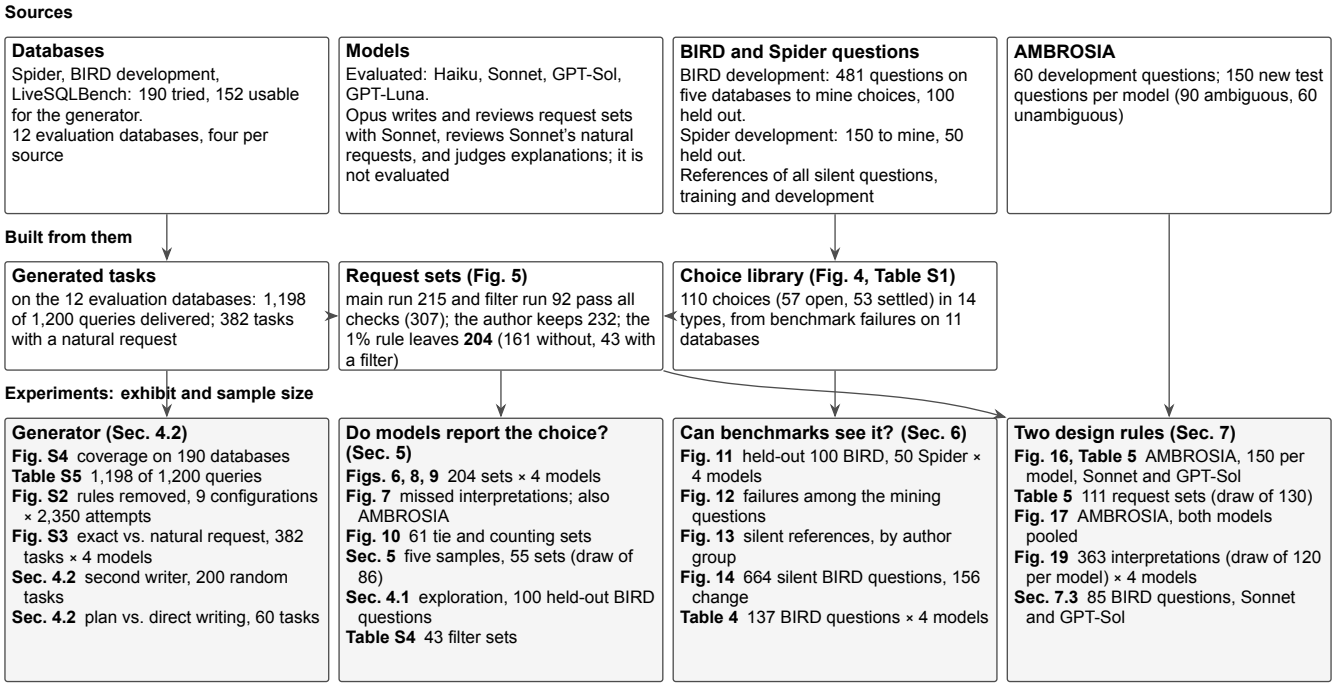


Figure S1: Data and experiments: sources (top), data built from them (middle), and the experiments with their sample sizes (bottom). Arrows show what is built from what and what each experiment uses.

a parent and are also connected by a link table, joined through the parent or through the link table. Which column a phrase means depends on the words, not the SQL, so the referent column has no count.

A lower bound. Each rule reads one SQL form of its choice, so it detects only part of the choices mined from failures: applied to the 85 library records of the twelve types, the rules find the choice in 50, including all ties and quantifier records, 27 of 34 counting records and 4 of 6 relationship records, but none of the aggregation, time scope or value matching records, whose forms (grouping by a non-key attribute, an average used as a threshold) the rules do not read. The counts are therefore a lower bound on how often the choices arise.

Review of the less common interpretations. Every reference that takes the less common interpretation was then read against fixed criteria (the SQL takes that interpretation, the wording leaves the choice open, and the other interpretation could change the answer), and every verdict is listed in the package for the author's review. Of 1,259 such references, 540 are dropped, most often because the wording settles the choice ("list the students with the highest", "how many different"), the join cannot repeat the counted entity, or a `LIKE` only escapes an apostrophe. Only this side was reviewed, so the shares in the table are conservative. The same drops apply to Section 6.1 of the paper.

Breadth. Table S2 counts every library choice by the benchmark whose failure revealed it, and every accepted request set by the source of its database. The library reflects where failures were diagnosed: its choices come from eleven databases, and several types were found in one benchmark only. The request sets show that the major types are not tied to those databases. Each of the five types with request sets beyond a single one occurs on at least eight of the twelve databases, and in databases from Spider, BIRD and LiveSQLBench.

Request sets. Table S1 counts the 204 accepted request sets by type, 43 of them with a choice in a filter (Appendix D). Over the 215 sets whose choice a model proposes, aggregation level contributes 63, ties 44, missing values 40, counting unit 35, calculation 15, relationship role nine, and time scope two. The remaining seven belong to the six additional labels: running-total interpretation (two), active-status filtering, comparison baseline, baseline population, baseline definition, and row-count versus quantity measurement (one each). These counts reflect the generation and selection procedure.

C CALCULATION CHOICES WITH REAL CONSEQUENCES

Table S3 surveys documented cases, across six domains, in which one open or mistaken choice of the kind our library records changed a consequential number or decision. None is a text-to-SQL failure. Each entry was checked against the original paper or official report; we left out cases whose primary source we could not reach, such as a bank's value-at-risk spreadsheet and a classic study of graduate admissions.

Table S2: Top: every library choice (Table S1), by the benchmark whose failures revealed it. Bottom: every accepted request set, by the source of its database. DBs: distinct databases.

Library choice type	Choices	DBs	Found in
Aggregation level	8	4	BIRD
Calculation	2	2	BIRD
Condition linkage	1	1	BIRD
Counting unit	34	4	BIRD, Spider
Missing values	11	5	BIRD, Spider
Negation	1	1	BIRD
Population scope	1	1	BIRD
Quantifier	3	1	Spider
Referent column	25	4	BIRD, Spider
Relationship role	6	2	BIRD, Spider
Strict cutoff	1	1	BIRD
Ties and selection	10	6	BIRD, Spider
Time scope	3	2	BIRD
Value matching	4	3	BIRD, Spider
Total	110	11	BIRD, Spider

Request set type	Sets	DBs	Spider	BIRD	LiveSQL
Aggregation level	61	11	20	16	25
Counting unit	30	8	12	1	17
Missing values	29	9	12	16	1
Negation	1	1	1	0	0
Strict cutoff	42	10	17	16	9
Ties and selection	41	11	9	12	20
Total	204	12	71	61	72

D REQUEST SETS WITH FILTER CONDITIONS

The filter sets of Section 2.4 of the paper come from one optional rule of the generator. It adds a single condition to the first stage of a query, on the rows of that stage's table, and builds the second interpretation at the same time. The five forms are: a date range, inclusive (>= and <=) or exclusive (> and <), with ends at the 30th and 70th percentile of a date column; a cutoff at a frequent value of a measure, `>= v` or `> v`; "without" a category, no record of the entity has it (`NOT IN`) or some record lacks it (`IN` with `<>`); two categories, on one record or on any records of the entity; and a per-entity count, over entities with matching rows (inner join) or over all entities (outer join, zero counts kept). The generator prefers forms it has used less often. The seed is 20261016; 161 of 180 requested queries are delivered, 121 of them have interpretations with different answers, and 92 request sets pass all checks.

Results. Table S4 gives the results per model on the 43 accepted filter sets. What models raise follows the wording of the open request. Forty of the 42 cutoff requests give a value "as the cutoff" or "threshold," with no comparison word, and the other two say "upwards of"; models return both the strict and the inclusive interpretation in 115 of the 159 cases where they compute both. The settled request is never called open. GPT-Luna returns both interpretations least often, as on the other sets.

Rejected sets. Section 4.2 of the paper explains why the author rejects every date-range set. Among the BIRD and Spider references whose question says "between" with no other cue, the one that excludes the ends is a Spider reference that applies strict bounds to the start and end columns of a staff role. A BIRD reference whose evidence says `BETWEEN 121 AND 130` but whose query excludes both ends is not counted, since its evidence settles the ends. Four further references use strict bounds moved one step out, or a negated range, and select the inclusive range. Of the 49 rejected sets, 17 are the date ranges, 15 are cutoffs whose wording ("passes", "clears", "on the high side of") already makes the comparison strict, five are "without" requests that already mean no such record, and one is a per-entity average whose request restricts the population. The other 11 test per-entity counts with or without unmatched entities (eight) and conditions on one or any records (three), choices whose other interpretation real references almost never take (Figure 4 of the paper).

E QUERY GENERATION DETAILS

Algorithm S1 lists the construction of a request set, Figure S2 shows what each construction rule contributes, and Figure S3 compares the exact description with the natural request. Figure S4 shows, for each feature, the share of 152 databases whose schema allows it and the share where a generated query with it was accepted by both model reviewers; and Table S5 the queries delivered on the 12 evaluation databases.

Formal statement. Let $\mathcal{L}(K_D)$ be the queries that the catalogue K_D of database D and the type rules allow, $\delta(p)$ a query's arithmetic and step depth, and $O(p)$ its operations. For target depths d and required operations R the generator samples from

$$\mathcal{P}(D, d, R) = \{p \in \mathcal{L}(K_D) : \delta(p) = d,\ R \subseteq O(p),\ I_D(p) = 1\}, \quad (1)$$

where $I_D(p) = 1$ means that all data checks pass. It keeps a partial query z only if some completion can still reach the target,

$$\mathrm{can}(z, d, R) = \mathbf{1}\{\exists (d, U) \in \mathrm{Reach}(z) : R \subseteq U\}, \quad (2)$$

where $\mathrm{Reach}(z)$ lists the depths and operation sets still possible. The look-ahead is exact for the rules, and no query is drawn twice. For a request set F with specific requests q_A, q_B and open request $q_\circ$, let $H_{ij}^{(r)} = 1$ when reviewer r accepts request i for interpretation j, $f_r = 1$ when r finds no changed or missing requirement, and $c(F) = 1$ when all code checks pass. The set is kept when

$$\mathrm{keep}(F) = c(F) \wedge \bigwedge_r \left[f_r \wedge H^{(r)}_{\{A,B\},\{A,B\}} = I_2 \right] \wedge \bigvee_r \left[H^{(r)}_{\circ A} \wedge H^{(r)}_{\circ B} \right]. \quad (3)$$

F SEEDS, VERSIONS AND SELECTION RULES

Seeds, versions and reruns. The grammar is `compose-v3.4`. Task requests, Spider splitting, request set selection, explanation allocation, and bootstrap resampling use seed 20261003; BIRD holdout selection uses 20261001; the paired wording comparison uses 20261004. Coverage uses seed 1 and depth-control runs seeds 1–3. Python was 3.11.5 and SQLite 3.41.2. During the experiments, the Claude command-line client changed from 2.1.287 to 2.1.289 and the Codex client from 0.147 to 0.160. Temporary usage-limit outputs were removed and retried rather than treated as SQL answers. A final query that failed under concurrent execution was rerun alone: ten such answers then matched their reference (nine GPT-Luna,

Table S3: Documented cases where one calculation choice changed a consequential number or decision, by domain, with the closest choice type in our library.

Domain	Case	Choice and consequence	Library type
Economics and policy	Public debt and growth [3]	Growth was averaged within each country before averaging across countries, so one year of New Zealand data weighed as much as decades of another country. With corrected data and weighting, high-debt countries grew 2.2% a year, not −0.1%.	aggregation level
Health care	Risk algorithm for extra care [6]	"Need" was computed as health-care cost. The algorithm looked accurate by standard measures, yet fixing the proxy would raise the share of Black patients receiving extra help from 17.7% to 46.5%.	which column a phrase means
Health care	Hospital mortality rates [7]	Four methods, given the same 2.5 million discharges, included 28–95% of them. 12 of 28 hospitals with higher-than-expected mortality by one method had lower-than-expected mortality by another.	population scope
Public health	1995 UK contraceptive pill warning [2]	A warning of higher thrombosis risk for two pill types was followed by pill use among under-16s falling from 40% to 27%, and by estimated NHS costs of £21 million for maternity care and £46 million for abortions. The author concludes that risk should be more carefully presented.	how a quantity is stated (not in our library)
Social science	Referees and skin tone [8]	29 teams analysed the same data. Their odds ratios ranged from 0.89 to 2.93; 20 teams found a significant effect and 9 did not, using 21 combinations of covariates.	several analysis choices
Science	Brain imaging [1]	70 teams analysed one dataset to test the same 9 hypotheses. No two chose the same workflow, and the results of the hypothesis tests varied substantially.	several analysis choices
Science	Gene names in spreadsheets [9]	Spreadsheet software converts gene names to dates and numbers by default; about one fifth of papers with supplementary gene lists in spreadsheets contain such conversions.	value representation
Engineering	Mars Climate Orbiter [5]	Thruster data in English units were used where the trajectory software expected metric units; the spacecraft was lost.	value representation
Engineering	Ariane 5 flight 501 [4]	A 64-bit floating-point value was converted to a 16-bit signed integer that could not hold it; the launcher broke up about 40 seconds after lift-off.	value representation

Table S4: Filter request sets (43 accepted), per model: both specific requests right (with exploration); among those, open requests answered with both interpretations; open requests that name the choice (Opus judge); specific requests right (single call); missed interpretations the model computes when asked, of all missed.

Model	Both right	Both returned	Named	Right	Missed, computable
Haiku	42	33	40	40	9 / 10
Sonnet	41	36	38	42	5 / 6
GPT-Sol	42	32	35	40	10 / 10
GPT-Luna	38	14	25	41	24 / 29

Table S5: Program construction on 12 databases, with no model calls after catalogue preparation. Every delivered SQL is distinct, has the requested depth, and passes both integrity checks.

Source	DBs	Delivered	Constructions	Seconds
Spider	4	398/400	289	127
BIRD	4	400/400	270	274
LiveSQLBench	4	400/400	309	37
Total	12	1198/1200	868	437

Source-level times are rounded individually; the total is rounded from the unrounded sum. Separate task review delivers 384/400 tasks, and language generation supplies 382/384 requests.

one GPT-Sol). Token counts are unavailable for the explanation runs, so latency is reported instead of cost.

Algorithm S1 Build a request set around one calculation choice

Require: Databases D, target depths d, required operations R
Ensure: Accepted request sets and reasons for rejected sets

for each requested task **do**
  Sample $p_A \in \mathcal{P}(D, d, R)$ with the look-ahead of Eq. (2)
  If no query is delivered, record the shortfall and continue
  Describe p_A; write and check its request
  Propose edits $p_A \to p_B$ that express a different choice
  Execute and describe both queries; require different answers
  Repair failed proposals once; review both interpretations
end for
Select one choice per task at most, balancing types and databases
for each selected choice **do**
  List what stays fixed and what changes
  Write requests specifying A, specifying B, and leaving it open
  Check requirements and all six request–interpretation pairs
  Revise a failing set once and recheck
  Keep F if keep(F); otherwise record the reason
end for

Rewrite comparison. The comparison of the original specific request (a) with the other texts was fixed before the run. The comparisons with the edited request (c) were added after an interim look at half of the interpretations.

Response selection. The explanation examples are the first saved responses displaying each tested pattern in file order. The worked

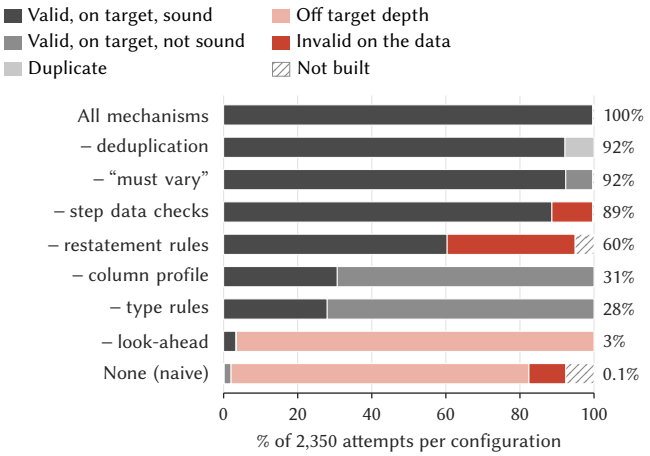


Figure S2: Generation outcomes with each construction rule removed (2,350 attempts). The labels give the share valid, on target and sound.

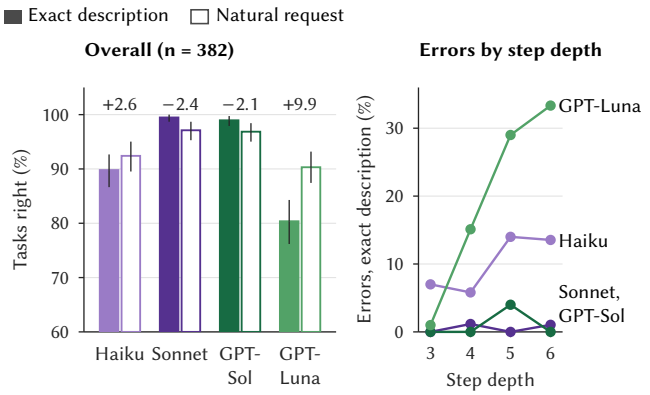


Figure S3: Exact description versus natural request on 382 generated tasks. Left: accuracy; numbers give the difference in points. Right: errors from the exact description by step depth.

aggregation example uses the smallest case-identifier hash among eligible request sets. Library examples use the first eligible record per type at high confidence, or medium confidence if none is high, with fixed benchmark and task ordering. Generated examples outside library labels enumerate retained cases before selecting the four shown in the main text. These selection rules support traceability; illustrative examples are not frequency estimates.

G EXAMPLES OF EACH CHOICE TYPE

Tables S6 and S7 show examples of each type, selected by a fixed rule rather than by hand. Library choices are taken from different databases, open choices before choices the wording settles, one from each benchmark when the type has both, shortest question first. Request sets are taken from different databases, one per source in turn, shortest open request first. Each example's question or request is given as the model received it. Answers says how the two

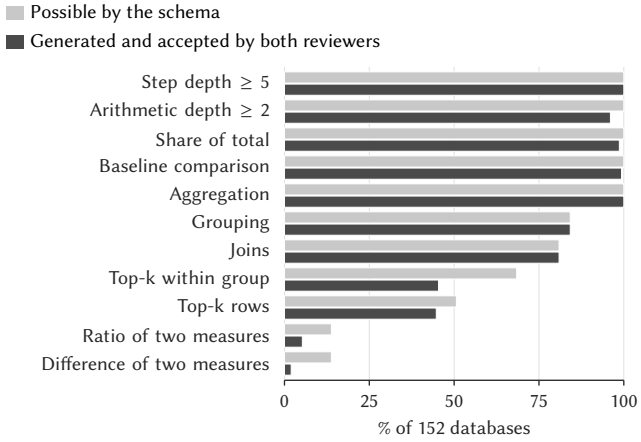


Figure S4: Features the generator can build, as a share of 152 databases: possible by the schema, and generated and accepted.

reference queries' results differ on the database. Models gives each model's answer to the open request in one call with explanation, in the order Haiku, Sonnet, GPT-Sol, GPT-Luna: B, both interpretations returned; N, one interpretation, with the choice named; S, one interpretation given silently, although the model computes each interpretation when asked; a dash, one interpretation, and the model does not compute both. The package holds every choice and every set, with its SQL.

Table S6: Selected library choices, up to 3 per type.

Source	Question	The choice	Wording
Aggregation level (3 of 8 shown)			
BIRD, `california_schools`	Which active district has the highest average score in Reading?	Whether to rank districts by one school's AvgScrRead value or by the average of AvgScrRead across each district's schools	open
BIRD, `financial`	Who are the account holder identification numbers whose who have transactions on the credit card with the amount is less than the average, in 1998? Evidence: Operation = 'VYBER KARTOU' refers to credit card withdrawal	Which transactions the 'average' amount threshold is computed over	open
BIRD, `thrombosis_prediction`	For all patient born in 1982, state if their albumin is within normal range. Evidence: Year(Birthday) = '1982'; albumin is within normal range refers to ALB between 3.5 and 5.5	Whether the normal/abnormal status is given once per laboratory record or once per patient born in 1982, since each patient can have many Laboratory rows	open
Calculation (2 of 2 shown)			
BIRD, `financial`	Provide the IDs and age of the client with high level credit card, which is eligible for loans. Evidence: the credit card is high-level refers to card.type = 'gold'; eligible for loans refers to disp.type = 'OWNER'	How to compute the client's age from birth_date: subtract the birth year from the current year, or count completed years of life.	open
BIRD, `thrombosis_prediction`	For all patients who are older than 55 years old, what is the percentage of female who has abnormal prothrombin time (PT)? Evidence: older than 55 years old = SUBTRACT(year(current_timestamp), year(Birthday)) > 55; abnormal prothrombin time (PT) refers to PT > = 14; percentage = DIVIDE(SUM(PT > = 14 AND SEX = 'F'), SUM(PT > = 14)) * 100; female refers to sex = 'F';	Which population the percentage is taken over (the denominator)	settles it
Condition linkage (1 of 1 shown)			
BIRD, `toxicology`	Calculate the percentage of carcinogenic molecules with triple bonded Hidrogen atoms. Evidence: hydrogen refers to element = 'h'; label = '+' mean molecules are carcinogenic; triple bond refers to bond_type = '#'; percentage = DIVIDE(SUM(label = '+'), COUNT(molecule_id)) * 100.0 where element = 'h' AND bond_type = '#';	How a hydrogen atom is tied to a triple bond: either the hydrogen is one of the two atoms in the '#' bond (linked through `connected`), or the hydrogen and some '#' bond just appear in the same molecule (linked only through molecule_id).	open
Counting unit (3 of 34 shown)			
BIRD, `thrombosis_prediction`	What is the average age of the male patient with high cholesterol? Evidence: average age = DIVIDE(SUM(SUBTRACT(YEAR(NOW()), YEAR(birthday))), COUNT(ID)); male patient refers to sex = 'M'; high cholesterol refers to `T-CHO` > = 250;	Whether the average is taken once per patient or once per matching lab record, since a patient can have several Laboratory rows with T-CHO >= 250	open
BIRD, `toxicology`	What is the least common element of all carcinogenic molecules? Evidence: label = '+' mean molecules are carcinogenic	What unit 'least common' is counted in: the number of carcinogenic molecules that contain the element, or the number of atoms of that element across all carcinogenic molecules	open
Spider, `student_transcripts_tracking`	How many different degrees are offered?	What counts as one degree: a distinct degree name (degree_summary_name) or each degree program row in Degree_Programs	open
Missing values (3 of 11 shown)			
BIRD, `california_schools`	What is the phone number of the school that has the highest average score in Math?	How to handle the top AvgScrMath row in satscores when it has no matching record in schools: drop it and take the best-scoring school that does have a record, or keep it as the answer even though its phone is missing.	open
Spider, `cre_Doc_Template_Mgt`	Show all template ids and number of documents using each template.	Whether templates that no document uses should appear in the output with a count of 0, or only templates that at least one document uses should be listed	open
Spider, `network_1`	Show the student IDs and numbers of friends corresponding to each.	Whether students who have no rows in Friend should be listed with a count of 0, or left out.	open
Negation (1 of 1 shown)			
BIRD, `toxicology`	What percentage of carcinogenic-type molecules does not contain fluorine? Evidence: label = '+' mean molecules are carcinogenic; contain fluorine refers to element = 'f'; percentage = DIVIDE(SUM(element = 'f') * 100, COUNT(molecule_id)) where label = '+'; Should consider the distinct atoms when counting;	What it means for a carcinogenic molecule to 'not contain fluorine': the molecule has at least one atom that is not fluorine, or the molecule has no fluorine atom at all.	settles it
Population scope (1 of 1 shown)			
BIRD, `thrombosis_prediction`	How many patients with an Ig G higher than normal? Evidence: Ig G higher than normal refers to IGG >= 2000; Should consider DISTINCT in the final result;	Whether the patient count only includes patients who also have records in the Examination table (because of the inner join to Examination), or every patient with any lab IgG result of at least 2000	settles it
Quantifier (1 of 3 shown)			
Spider, `world_1`	What are the countries that have greater surface area than any country in Europe?	Whether 'greater than any country in Europe' means larger than at least one European country (compare to the smallest European area) or larger than every European country (compare to the largest European area)	open
Referent column (3 of 25 shown)			
BIRD, `california_schools`	Please specify all of the schools and their related mailing zip codes that are under Avetik Atoian's administration.	Which administrator slots count as being 'under Avetik Atoian's administration': only the primary administrator columns (AdmFName1/AdmLName1), or any of the three administrator slots.	open
BIRD, `financial`	What year was account owner number 130 born?	Which ID the number 130 in 'account owner number 130' refers to: the account ID or the client ID	open
Spider, `car_1`	What are the makers and models?	Which column supplies 'makers': the integer foreign key model_list.Maker, or the maker name car_makers.Maker reached by joining on that key	open
Relationship role (2 of 6 shown)			
BIRD, `toxicology`	Which molecule consisted of Sulphur atom with double bond? Evidence: sulphur refers to element - 's'; double bond refers to bond_type = ' = ';	Does the sulfur atom itself have to be part of the double bond, or is it enough that the molecule has both a sulfur atom and a double bond somewhere?	open
Spider, `network_1`	What is the lowest grade of students who do not have any friends?	Whether a student who appears in Friend only as friend_id (someone else lists them as a friend) counts as having a friend	open
Strict cutoff (1 of 1 shown)			

Source	Question	The choice	Wording
BIRD, `thrombosis_prediction`	Provide the ID and age of patient with lactate dehydrogenase (LDH) between 100-300 index above the normal range. Evidence: age refers to SUBTRACT(year(current_timestamp), year(Birthday)); lactate dehydrogenase (LDH) between 100-300 index above the normal range refers to LDH between 600 and 800;	Whether the LDH range endpoints 600 and 800 are included (strict vs. inclusive bounds)	open
Ties and selection (3 of 10 shown)			
BIRD, `superhero`	Give the hero ID of superhero with the lowest attribute value. Evidence: lowest attribute value refers to MIN(attribute_value);	How to handle ties at the minimum attribute_value: return every tied hero, or return just one hero as the singular 'the hero ID' suggests.	open
BIRD, `thrombosis_prediction`	For the patient who has the highest Ig A within the normal range, what is his or her diagnosis? Evidence: highest Ig A within the normal range refers to MAX(IGA BETWEEN 80 AND 500);	How to handle several lab records tied at the maximum in-range IgA value (IGA = 500): return one row or every tied patient's diagnosis.	open
Spider, `world_1`	Give the name of the nation that uses the greatest amount of languages.	Several countries tie for the most languages (12). The question does not say which of them to return, so the answer depends on how the tie is broken.	settles it
Time scope (2 of 3 shown)			
BIRD, `financial`	For the client who applied the biggest loan, what was his/her first amount of transaction after opened the account?	Whether a transaction on the account-opening date counts as 'after opened the account' (inclusive vs. strictly later boundary)	open
BIRD, `thrombosis_prediction`	For all patients with normal uric acid (UA), what is the average UA index based on their latest laboratory examination result? Evidence: uric acid (UA) with normal range refers to UA < 8.0 and SEX = 'M' or UA < 6.5 and SEX = 'F'; average UA index refers to AVG(UA)	Which date counts as the 'latest laboratory examination': the single latest date in the whole Laboratory table, or each patient's own latest test date	settles it
Value matching (3 of 4 shown)			
BIRD, `financial`	How much, in total, did client number 617 pay for all of the transactions in 1998?	Which 1998 transactions count as the client having 'paid': every transaction on the client's accounts, or only outgoing ones (debits).	open
BIRD, `thrombosis_prediction`	Please list the disease names of the patients that have a proteinuria level higher than normal. Evidence: disease names refers to Diagnosis; proteinuria level higher than normal refers to `U-PRO` > = 30;	Whether the threshold `U-PRO` >= 30 compares the TEXT column `U-PRO` as text (string order) or as numbers	settles it
Spider, `car_1`	Which model saves the most gasoline? That is to say, have the maximum miles per gallon.	Whether the TEXT-typed MPG column is ranked by its stored string value or by its numeric value when picking the maximum	settles it

Table S7: Selected accepted request sets, up to 4 per type.

Set	Open request	Interpretations	Answers	Models
Aggregation level (4 of 61 shown)				
`chinook_1` `1adc54e3cf`	Which media types (by name, counting only tracks that have a media type) have a bytes-per-millisecond rate higher than the simple average of that rate across all media types? For each, show the media type name, its rate, and how much it exceeds that average, unrounded.	**A:** Each media type's rate is the mean of its tracks' individual Bytes/Milliseconds ratios; list the media types whose rate is above the unweighted mean of all media types' rates, with the rate and how far above it is. **B:** Each media type's rate is its total Bytes divided by its total Milliseconds; list the media types whose rate is above the unweighted mean of all media types' rates, with the rate and how far above it is.	value 181.7 vs 179.5	`-BBN`
`formula_1` `c759056a31`	For each circuit country, what are the average constructor standings points and wins (the season-to-date figures) at races held there? Also, how far is each country's average wins above or below the average of the country figures?	**A:** A country's averages are the mean of per-race averages, so each race counts equally. **B:** A country's averages are the plain mean over all of its standings records pooled together.	value 0.039 vs -0.03776	`S--S`
`thrombosis_prediction` `7fd142e2d9`	By patient sex, what are the average aCL IgG and ANA values in the examination data (only examinations linked to a patient, missing values ignored)? Also give each sex's average ANA as a ratio of the mean of the per-sex ANA averages, unrounded.	**A:** Average aCL IgG and ANA over all matched examination records within each sex. Each examination counts once. **B:** Average each patient's examinations into per-patient values first. Then average those values within each sex, so each patient counts once.	value 3.281 vs 3.334	`SSSS`
`alien` `cde68b2348`	For observatory stations whose average telescope bandwidth usage (`bandusagepct`) is above average, what is their mean bandwidth usage, and how far above average are they on average? Ignore missing usage values, and give both figures unrounded in a single row.	**A:** Compare each station's average bandwidth usage with the unweighted mean of the station averages. **B:** Compare each station's average bandwidth usage with the mean bandwidth usage over all individual telescopes.	value 24.9 vs 25.03	`SBBN`
Counting unit (4 of 30 shown)				
`baseball_1` `00d645c60b`	How do manager award winners break down by batting hand (`bats`)? Use only awards that match a record in `player`, and treat a blank or missing hand as its own group. For each hand, show the hand, how many manager award winners it has, and that figure's share of the overall total (no rounding).	**A:** For each batting hand, count the manager award records won by people with that hand, and give each hand's share of all matched award records. **B:** For each batting hand, count the distinct managers with that hand who won at least one award, and give each hand's share of all such managers.	value 122 vs 55	`BBBB`
`sakila_1` `fc16192583`	For each actor, show their actor ID, how many films they're credited in, that count as a fraction of the total, and how far that fraction is above or below the average fraction across actors. Return the top 10 by that difference, highest first, unrounded.	**A:** Each actor's film count is divided by the total number of casting credits in film_actor, which gives the actor's share of all credits. **B:** Each actor's film count is divided by the number of distinct films that have a cast in film_actor, which gives the fraction of those films the actor appears in.	value 0.001591 vs 0.008716	`-N--`
`formula_1` `f27aed7e53`	Using driver standings linked to their races by race id (leaving out standings with no matching race), for each race name (`races.name`) get the average season-to-date championship points (`driverStandings.points`, ignoring missing values) and how big its standings field is. Then focus on the race names whose field size is strictly above the average field size across all race names. In one row, give the average of their average points, their average field size, and how far above the overall average their field size is on average. No rounding.	**A:** Per race name, count every driver standings record. Compare those counts with their average across names and summarise the names that are above average. **B:** Per race name, count distinct drivers in the standings. Compare those counts with their average across names and summarise the names that are above average.	1 rows; 1 differ	`SBSS`

Set	Open request	Interpretations	Answers	Models
insider e6f24d2438	For each trader type, how many analysed trade records are there in the advanced behaviour data (only where the trade and its trader can be found), and what share of the overall total does each type make up? Keep the share unrounded.	**A:** For each trader type, count the advanced behaviour analysis records that join to a transaction and a trader. Report that count and its share of the total count over all trader types. **B:** For each trader type, count the distinct transactions that have at least one advanced behaviour analysis and a matching trader. Report that count and its share of the total count over all trader types.	value 242 vs 241	BBBS
Missing values (4 of 29 shown)				
baseball_1 f74c2bad9d	For players who received award votes (matched to player by player_id), what is the average points_won for each death country (ignoring missing points)? Show each death country, its average, and how far it is from the simple average of those per-country averages, without rounding.	**A:** Average the points won per death country, with a blank death country kept as its own group. Compare each group's average with the plain average of all group averages, the blank group included. **B:** Average the points won per death country, leaving out players with no recorded death country. Compare each country's average with the plain average of the remaining countries' averages.	8 vs 7 rows	NN-N
formula_1 af91f25689	Which 10 drivers have the most qualifying entries compared with the average driver? For each, show the driver ID, how many qualifying records they have, and that number divided by the average number of qualifying records per driver (unrounded), highest ratio first.	**A:** The average number of qualifying records per driver is computed only over drivers who appear in the qualifying table. **B:** The average number of qualifying records per driver is computed over every driver in the drivers table, with drivers who never qualified counting as zero.	value 4.112 vs 22.63	SBBB
student_club 55be105b98	Group members by their major (link_to_major) and show each major id, how many members it has, and that count's share of the total. List the top 10 by share, highest first, with no rounding.	**A:** Members without a major form their own group, and they are included in the total used as the denominator. **B:** Members without a major are left out of both the groups and the total used as the denominator.	10 rows; 1 differ	BBBN
crypto b61f4a52bd	For each order in orderexecutions.ordersmark (executions with no order reference grouped together), what's the average size (fillcount) of the fills it got? Keep the orders whose average beats the average across all orders, and note by how much. Then, among those, keep the ones that also beat that group's own average, and note that margin too. Give me one summary row with three values: the average per-order fill size of the final orders, their average margin over the first benchmark, and their average margin over the second. Leave everything unrounded.	**A:** Zero-quantity executions count as fills of 0 in each order's average fill quantity. **B:** Zero-quantity executions are not fills and are left out of each order's average fill quantity.	value 4.024 vs 4.021	SSSS
Negation (1 of 1 shown)				
customers_and_ invoices 2e529bb193 (filter)	Ignoring transactions with no invoice number, which invoice numbers have transactions without the type 'Payment' (exact text)? Show each invoice number with the total number of transactions recorded for it. Sort by that count from highest to lowest and give the top 10.	**A:** Invoices with no Payment transaction at all. For each one, count its transactions and show the top 10 by count. **B:** Invoices with at least one transaction that is not a Payment. For each one, count all its transactions (Payments included) and show the top 10 by count.	4 vs 7 rows	SSNS
Strict cutoff (4 of 42 shown)				
chinook_1 06934ae88f (filter)	Using 3.98 as the cutoff on the invoice total (Invoice.Total) to keep only the higher-value invoices, what is their average total, as a single unrounded value? Ignore missing totals, and give a missing result if no invoices qualify.	**A:** Invoices whose total equals 3.98 are included in the average. **B:** Only invoices whose total is strictly greater than 3.98 are included in the average.	1 rows; 1 differ	BBBS
financial 8a55588ad2 (filter)	Applying a cutoff of 2802.0 to the debited amount (order.amount), what is the average debited amount of the large standing orders, unrounded?	**A:** Orders of exactly 2802.0 are included. **B:** Only orders above 2802.0 are included.	1 rows; 1 differ	BBBB
formula_1 57277c47b4 (filter)	Among driver standings entries where the driver has upwards of 1 point so far that season, what is the average season-to-date win count? Ignore missing values and don't round.	**A:** Average the season-to-date win counts over standings entries whose season-to-date points are 1 or more. **B:** Average the season-to-date win counts over standings entries whose season-to-date points are more than 1.	1 rows; 1 differ	BBBS
insider 8ed79619ea (filter)	Using 0.149 as the cutoff, how many advanced behaviour records have high peer correlation (peercorr)? Give a single count.	**A:** Peer correlation values equal to 0.149 are counted (at least 0.149). **B:** Only peer correlation values strictly greater than 0.149 are counted (more than 0.149).	1 rows; 1 differ	BBBN
Ties and selection (4 of 41 shown)				
sakila_1 b77a784ee8	What are the top 10 films by number of copies in our inventory? For each, show the film_id, how many copies we have, and what share of all inventory copies that is (copies divided by the total number of copies, unrounded), with the highest share first.	**A:** List the top 10 films by share of inventory copies, returning exactly 10 rows and choosing arbitrarily among films tied at the cut. **B:** List the top films by share of inventory copies, keeping every film whose copy count reaches the 10th-highest count, so ties at the cut are all included.	10 vs 72 rows	-N--
financial 53afca1d49	What are the top 10 branch districts (account.district_id) by share of all accounts? For each, show the district id, its number of accounts and its share of all accounts, without rounding, listed from the highest share down.	**A:** Rank districts by their share of accounts and return exactly the first 10 rows, dropping tied districts arbitrarily at the cut. **B:** Rank districts by their share of accounts and return every district whose share reaches the 10th-highest share, including all ties at the cut.	10 vs 12 rows	SNSS
formula_1 c1a8ce4d18	Which circuit cities have an above-average share of qualifying? For each, show the city, its qualifying count, that count as a share of the total across all cities, and how far its share is above the average city share (all unrounded).	**A:** Each city's count is its number of driver qualifying entries, and its share is its share of all qualifying entries. **B:** Each city's count is its number of distinct races with qualifying data (qualifying sessions), and its share is its share of all such sessions.	16 vs 17 rows	SS--
alien ee6b268047	What are the top 10 telescopes by share of detected signals? For each telescope reference (telescref), show how many signals it detected and what fraction of all signals that is (unrounded), from the largest share down.	**A:** Rank telescopes by their share of all signals and return exactly 10 rows, choosing arbitrarily among telescopes tied at the 10th place. **B:** Rank telescopes by their share of all signals and return every telescope whose share is at least the 10th-highest share, including all ties at the cut.	10 vs 51 rows	-NS-

REFERENCES

[1] Rotem Botvinik-Nezer, Felix Holzmeister, Colin F. Camerer, et al. 2020. Variability in the Analysis of a Single Neuroimaging Dataset by Many Teams. *Nature* 582 (2020), 84–88.

[2] Ann Furedi. 1999. The Public Health Implications of the 1995 'Pill Scare'. *Human Reproduction Update* 5, 6 (1999), 621–626.

[3] Thomas Herndon, Michael Ash, and Robert Pollin. 2014. Does High Public Debt Consistently Stifle Economic Growth? A Critique of Reinhart and Rogoff. *Cambridge Journal of Economics* 38, 2 (2014), 257–279.

[4] Jacques-Louis Lions et al. 1996. *ARIANE 5 Flight 501 Failure: Report by the Inquiry Board.* Technical Report. European Space Agency and Centre National d'Études Spatiales, Paris.

[5] Mars Climate Orbiter Mishap Investigation Board. 1999. *Mars Climate Orbiter Mishap Investigation Board Phase I Report.* Technical Report. National Aeronautics and Space Administration.

[6] Ziad Obermeyer, Brian Powers, Christine Vogeli, and Sendhil Mullainathan. 2019. Dissecting Racial Bias in an Algorithm Used to Manage the Health of Populations. *Science* 366, 6464 (2019), 447–453.

[7] David M. Shahian, Robert E. Wolf, Lisa I. Iezzoni, Leslie Kirle, and Sharon-Lise T. Normand. 2010. Variability in the Measurement of Hospital-Wide Mortality Rates. *New England Journal of Medicine* 363, 26 (2010), 2530–2539.

[8] Raphael Silberzahn, Eric L. Uhlmann, Daniel P. Martin, et al. 2018. Many Analysts, One Data Set: Making Transparent How Variations in Analytic Choices Affect Results. *Advances in Methods and Practices in Psychological Science* 1, 3 (2018), 337–356.

[9] Mark Ziemann, Yotam Eren, and Assam El-Osta. 2016. Gene Name Errors Are Widespread in the Scientific Literature. *Genome Biology* 17 (2016), 177.